

Quiz 3

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (10 points) Evaluate the following integrals:

(a) $\int \sin^5 \theta \, d\theta$

$$\begin{aligned} \text{let } u = \cos \theta, \, du &= -\sin \theta \, d\theta \\ \int \sin^5 \theta \, d\theta &= \int \sin^4 \theta \sin \theta \, d\theta \\ &= \int (1 - \cos^2 \theta)^2 \sin \theta \, d\theta \\ &= \int (1 - u^2)^2 (-du) \\ &= \int -(1 - 2u^2 + u^4) \, du \\ &= -u + \frac{2}{3}u^3 - \frac{1}{5}u^5 + C \\ &= -\cos \theta + \frac{2}{3}\cos^3 \theta - \frac{1}{5}\cos^5 \theta + C \end{aligned}$$

(b) $\int \tan x \sec^4 x \, dx$

$$\begin{aligned} \text{let } u = \sec x, \, du &= \sec x \tan x \, dx \\ \int \tan x \sec^4 x \, dx &= \int \sec^3 x (\sec x \tan x) \, dx \\ &= \int u^3 \, du. \quad \left\{ \begin{array}{l} \text{or let } u = \tan x, \\ du = \sec^2 x \, dx \end{array} \right. \\ &= \frac{u^4}{4} + C \\ &= \frac{\sec^4 x}{4} + C \quad \left\{ \begin{array}{l} \int \tan x \sec^4 x \, dx \\ = \int \tan x \sec^2 x \sec^2 x \, dx \\ = \int u(1+u^2) \, du \\ = \int u + u^3 \, du \\ = \frac{u^2}{2} + \frac{u^4}{4} + C \\ = \frac{\tan^2 x}{2} + \frac{\tan^4 x}{4} + C \end{array} \right. \\ &\quad \text{equivalent!} \end{aligned}$$

(c) $\int_0^{\pi/2} \sqrt{\frac{1 - \cos x}{2}} \, dx$

$$\begin{aligned} &\int_0^{\pi/2} \sqrt{\frac{1 - \cos x}{2}} \, dx \\ &= \int_0^{\pi/2} \sqrt{\sin^2 \frac{x}{2}} \, dx \\ &= \int_0^{\pi/2} \sin \frac{x}{2} \, dx \\ &= \left[-2 \cos \frac{x}{2} \right]_{x=0}^{x=\pi/2} \\ &= -2 \left[\cos \frac{\pi}{4} - \cos 0 \right] \\ &= -2 \left(\frac{\sqrt{2}}{2} - 1 \right) \\ &= 2 - \sqrt{2} \end{aligned}$$

(d) $\int \frac{1}{x^2 \sqrt{16 - x^2}} \, dx.$

$$\begin{aligned} \text{let } x &= 4 \sin \theta, \text{ then } dx = 4 \cos \theta \, d\theta \\ \text{and } \sqrt{16 - x^2} &= 4 \cos \theta \\ \int \frac{1}{x^2 \sqrt{16 - x^2}} \, dx &= \int \frac{1}{(4 \sin \theta)^2 4 \cos \theta} 4 \cos \theta \, d\theta \end{aligned}$$

$$\begin{aligned} &= \int \frac{1}{16 \sin^2 \theta} \, d\theta \\ &= \frac{1}{16} \int \csc^2 \theta \, d\theta \\ &= -\frac{1}{16} \cot \theta + C \\ &= -\frac{1}{16} \frac{\sqrt{16 - x^2}}{x} + C \end{aligned}$$

$\Rightarrow \sin \theta = \frac{x}{4}$
 $\Rightarrow \cot \theta = \frac{\sqrt{16 - x^2}}{x}.$